COL7378: Principles of Autonomous Systems

Semester II, 2025-26

Course Organization

Rohan Paul

Course Information

- Name
  - Couse Name: Principles of Autonomous Systems
  - Credits: 4
  - LTP: 3-0-2
- Timing
  - Slot: AA
  - Class Days: Mondays - Thursdays
  - Class Times: 2:00pm – 3:20pm
  - Room: LH 605.

Course Logistics

- Instructor
  - Prof. Rohan Paul
  - Email: rohan@cse.iitd.ac.in
  - Course webpage: https://www.cse.iitd.ac.in/~rohanpaul/teaching
- Teaching Assistant
  - Allocation is still in progress (will be notified in due course).
- Communication
  - Formal communication will be on the course mailing list by the institute.
  - Piazza forum will be setup to discuss any assignment related queries.
  - Theory-related or other doubts please ask in the class so that everyone benefits. You may also ask after class lectures.

Who should take this course?

- Pre-requisites
  - Foundation course in AI/ML
    - Pre-requisites: COL333 / COL774 / COL3333/ COL7333/ COL7341/ AIL7025/ AIL7024.
    - Permitted: AIL7022 / APL745 / APL744 / APL747 / ELL 409.
    - The offering of COL864 (Sem 1 2025-26) course last semester will have a >30% overlap with COL7378/COL778 course (Sem 2 2025-26) this semester. Hence, who have done COL864 last semester are requested <u>not</u> to register for COL7378 this semester. Instead, please explore other courses.
  - PG Students
    - PhD/MS(R)/M. Tech. students must have taken one of the above-mentioned courses at IIT. Prior AI courses done online or in UG will not be considered (it creates heterogeneity in class).
    - Note: background courses are necessary even if you are using such tools in your research. Further, needing to use the tools for your thesis is not grounds for enrolment.
  - No sit throughs
    - Course participation is only for registered students. <u>No sit throughs are permitted</u> as they create heterogeneity in the class.
  - Others
    - Note: work done in internships/clubs will not merit enrolment in the course. Background courses are necessary.

Who should take this course?

- Other Skills
  - Programming proficiency
    - Python and ability to install/work with external libraries/install and use a simulator on your own.
  - Foundational knowledge
    - Probabilistic models, basic deep learning, basic search algorithms, Markov Decision Processes.
  - Prior experience
    - Training neural networks using PyTorch etc. will be an advantage.
    - A tutorial will be provided in the course for the concerned assignments.

Evaluation Criteria (tentative)

- Component I: Examination (theoretical aspects): 60%
  - Mid-term and Major Exam
- Component II: Assignments (practical aspects): 40%
  - Implementation and experimentation. Typically (approx. 3-4 in number)
- Margin: ~15% (discretionary)
  - Instructor's discretion as contingency incase any adjustments need to be done during the semester.
- Note:
  - There may be a small quiz component (will be notified in advance).
  - For the weightage of course sub-components a mean and a small approx. range may be provided. The final weightages will be notified after all sub-components finish.

Other Policies

- Pass Criteria for Credit
  - 30% of the total score in the course
- Criteria for Audit Pass (NP)
  - Grade "B-" or higher AND at least 30% in each course component (the student must pass the examination and assignment components individually).

Teaching Mode

- Lectures
  - Link for slides will appear on the course web page. Slides will be typically updated after the class. Blackboard teaching may supplement the slides.
  - Lectures will often contain insights and examples from experience. Audio-visual material as well as recent papers/articles may also be shared in class.
  - The lecture material has evolved over the years with student feedback and developments in the field.
  - Extra teaching days may be utilized and will be announced in advance, typically one class earlier (likely timings are Wednesday 12pm or Saturday 10am).
  - Class participation is highly encouraged. Please ask any class-related doubts in the class or after class.

Class Attendance

- Attendance
  - Class attendance will be taken, and an official record will be maintained.
  - The record may be shared with the dept./institute as per policy.
- Class Decorum
  - Please maintain decorum in class. Please do not engage in movement/side conversations which can affect other students concentrating in class.

Examinations

- Format
  - Standard offline mode.
  - Information regarding exam logistics will be provided before the exam.
- Exam material
  - Regularly reviewing the class material is highly recommended. This will help internalize the concepts and likely improve performance in the assignments. Please do not query for previous years questions papers.

Assignments

- Relevant instructions will be provided for each assignment.
- Assignments will be individual or in pairs and will be notified when the assignment is released.
  - In case of partners, both are expected to contribute relatively equally and be familiar with each others' part.
  - Please discuss actively with your partner so that both students learn the material well. Unequal marks may be provided in case of a large gradient in effort/understanding.
- Preparation
  - Basic proficiency in Python/C++ and ability to install/work with external libraries if required for an assignment on your own.
  - Any starting code will be tested on Linux systems. Students are expected to resolve building on any OS on their own. A linux-based system will be used for assignment evaluation.

Assignments

- Submissions
  - Typical assignment submission time will around 5pm.
  - Ensure access to Moodle.
  - Any delays beyond the submission time will result in late penalty of X% per day (from the submission day and time) up to generally 1-2 days. Typically, X is 10%.
  - No deadline extensions are permitted (1-2 late days with adjustment are available).
  - Any changes to the code or replacement of files after deadline are not permitted.
- Other Guidelines
  - Please carefully follow the guidelines and input/output formats if provided in your code. Not following the guidelines stall evaluation and take away time which is better spent on teaching. Hence, a lowering of scores (adjustment) will be done to be fair to other students who follow the guidelines.
  - Viva may be conducted for an assignment. The schedule will be notified in advance.
  - Please do not ask for a late viva or submitting an assignment after closure.
- Please start the assignments on time. Please manage your different courses well.

Academic Integrity

- Please listen to your conscience. Please do not cheat.
- Please only submit the assignment only based on your own effort and understanding.
  - Please write code or other written submissions from scratch independently. Sharing of code or parts of it or posting it online will constitute a violation of the honor code.
  - Do not look at or refer to code written by anyone else. You may discuss the problem; however the code implementation must be original. Discussion will not be grounds to justify software plagiarism.
  - Do not submit code copied from the internet, obtained via Generative AI tools or copied from students (current or past). Such solutions will be part of the plagiarism checking.
  - If the assignment requires a report to be submitted, please write on your own based on your own understanding. Copying in the report will also constitute as plagiarism.
  - Any mechanisms such as code similarity s/w or others can be used by the department to assess plagiarism.
- For fairness to honest students, plagiarism cases will be penalized with exclusion from assessment and additional penalties.

Honor Code Violations

- We are duty bound to follow the disciplinary procedures of the Department and the Institute in this regard.
  - Plagiarism in assignment/exam will result in zero in the assignment/exam and an additional penalty on an absolute scale (at least -10 absolute). Department and institute procedures such as DISCO and an F-grade will follow.
  - Copying or cheating in even a sub-part of an assignment or an exam will be counted as plagiarism in the whole assignment/exam. The whole assignment and exam will be made void with additional penalties and Dept. procedures.
  - In case an assignment allows working in pairs, both students will receive the same penalty even if only one student may be involved.
- Institute procedure will be followed
  - Dealing with plagiarism is part of the instructor's and the department's mandate. Please do not argue if you are found to be plagiarizing.
  - Any discussions will be at the department and institute committee level. We will forward the names and the findings as per institute guidelines. Information will be shared with other faculty members for future courses and projects.

Other Emergencies

- Any requests for late submissions or re-appearing in an exam on the grounds of medical emergencies must be accompanied with a medical certificate from a qualified doctor indicating that you were unwell in the period of submission and a proof of prescription before the due date of the assessment.
- Any request to appear in the re-examination will require formal proof that you were unwell or exceptional circumstances prevailed during the scheduled examination due to which you could not appear.
- Please follow institute guidelines for missing examinations.

Teaching Assistants

- <u>Please be respectful to the teaching assistants.</u>
- Teaching Assistants play a very important role in the course. They do a lot of work behind the scenes to ensure the success of the course.
- Please do not send personal messages on phone/Whatsapp or Teams. Please only use official channel for communication (Piazza).
- TAs are students who are contributing to the course as TAs in their official capacity.
- Some of you will also become TAs in the future!

What is this course about?

- **Autonomous Systems**
  - *We will study the foundations of a class of AI systems called Embodied AI or Physical AI. AI-based robots are also included.*
  - **What is unique about them?**
    - Sensing, decision making and interacting with the physical world.
    - Physical interaction, exchange of energy with the physical environment.
    - In essence, this is AI that interacts with the physical world or with humans
  - **Why now?**
    - Advances in computing and sensing is making computing + AI systems pervasive.
    - We can start imagining a future where we can interact with such systems as we can now interact with other hardware.
    - AI will leave the computer and start becoming a part of the physical world.

Topics

Overview

- Course Organization/Introduction
- Agent Representation
- State Estimation
- Decision-making under uncertainty
- Task Planning
- Motion Planning
- (Deep) Reinforcement Learning in Robotics/Imitation Learning
- Online Planning (MCTS)
- Partially-observed MDPs
- Topics in Perception/Generative AI

Develop an algorithmic toolkit

- Estimation
  - How to make decisions when the state is not known?
  - Discrete and continuous spaces.
- Planning
  - How to decide actions towards a long-term goal.
  - Various resolutions of planning
- Learning
  - How to use experience to make better decisions.
  - Using human demonstrations or exploration to learn a policy.
- Practical
  - How to practically realize/model such systems.

The topic list is tentative and will be updated as the lectures are presented. The order may also change to some extent. Please see the previous offerings of the course to get a sense of the topics. These will be updated in this offering based on the current state of the field and feedback from past students.

Please Review Before Class

- Please review the following specific topics <u>before</u> class as pre-requisites.
  - Classical search: A* Search
    - Fundamentals of planning
  - Markov Decision Processes
    - This is a model of agent behaviour and needed for RL.
  - Basics of Bayesian Networks
    - Uncertainty in actions and state is modeled via BayesNets - need for state estimation.
  - Deep Learning
    - Basics of neural networks and training. We will combine these tools with decision making.
- Please refer to a standard AI textbook to refresh the above.

Books and References

- Primary resource is the material covered in class.
- Books
  - Material is derived from various sources. The relevant sections will be indicated on the course webpage.
    - Artificial Intelligence: A Modern Approach (3rd Edition). Russell, Stuart J., and Peter Norvig. Link.
    - Sebastian Thrun, Wolfram Burgard and Dieter Fox. Probabilistic Robotics. MIT Press, 2005.
    - Reinforcement Learning (Second Edition). Richard Sutton and Andrew Barto. MIT Press. 2018. Online.
    - Mykel J. Kochenderfer, Decision Making Under Uncertainty: Theory and Application, MIT Press, 2015
    - Simon J. D. Prince, Understanding Deep Learning, MIT Press, 2023.
- Paper references
  - Any paper references mentioned will be provided on the course webpage.

Learning Objectives

- At the end of the course
  - Students will understand the foundations of building learning AI agents that can sense—plan-act in the real world.
- AI skills
  - Models of state estimation, planning, learning by interaction.
  - A strong toolkit for building systems that take decisions.
- Applicability
  - Settings where the AI system is interacting with the world and making intelligent decisions.
  - Academia – Physical AI is an important component of AI. Test bed for algorithms.
  - Industry I - Autonomous system, Robotics and CS applications.
  - Industry II – Generic industry (finance, intelligent machines, HRI systems, recommender systems) that involve sequential decision making and human interaction.

Other Information

- This course will focus on agentic view of AI (autonomous systems)
  - A robotic system (ground/air vehicle or manipulator) will be modeled as an AI agent capable of sensing and taking simple actions in the environment.
  - The detail control and physical aspects of the system will be abstracted to a certain degree for the models discussed in the course.
- Beyond the course
  - Prepares the student to undertake independent project work in this area in future.
  - For projects, please discuss after the course. Project possibilities are based on performance, conduct and availability. Course does not guarantee subsequent projects.
  - Also, please do not ask for recommendation letters based solely on this course. I consider recommendation letters only for substantial research-oriented engagement.

Next Time

- This Class
  - Course Organization
- Next Class
  - Introduction